

PROJECT VIOLET MILK

Frontend Development Map & UI Organs Blueprint

TARGET PLATFORM	Web Application (React 18 / Next.js)
TARGET USER	Indian Law Enforcement Officers (IOs) & Cyber Crime Units
COMPLIANCE STANDARD	Bharatiya Sakshya Adhinyam (BSA) 2023 & BNSS 2023
STATUS	Court-Ready Prototype Architecture

COURT-READY PROTOTYPE ARCHITECTURE

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Target Platform: Web Application (React 18 / Next.js)

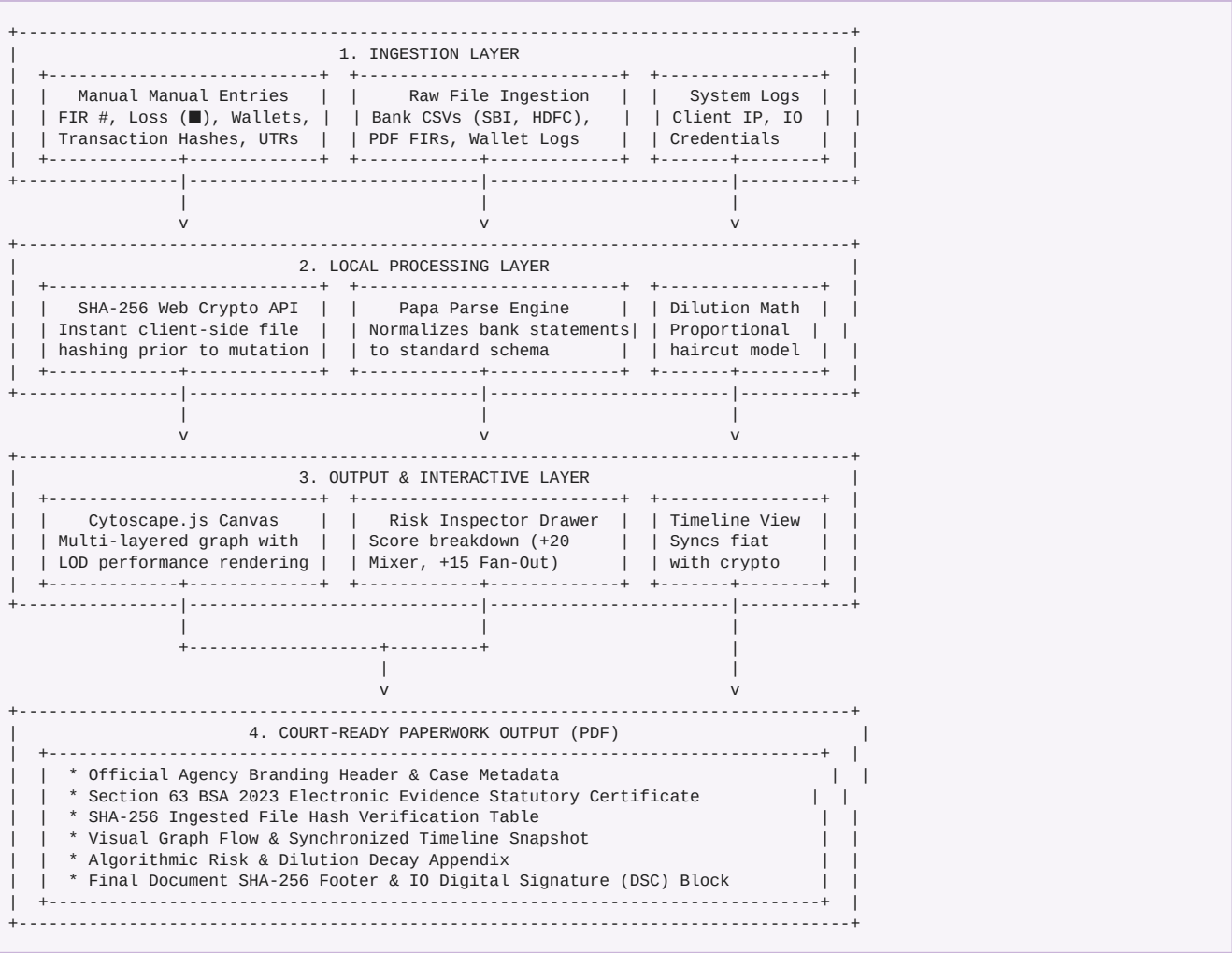
Target User: Indian Law Enforcement Officers (IOs) & Cyber Crime Units

Compliance Standard: Bharatiya Sakshya Adhiniyam (BSA) 2023 & BNSS 2023

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SECTION 1: FRONTEND DEVELOPMENT & DATA FLOW MAP

The following diagram maps the flow of data from raw police intake to court-admissible evidence output.



SECTION 2: UI ORGAN SPECIFICATIONS

ORGAN 0: Command Center (Executive Dashboard)

Purpose: Provide a tactical, single-pane control room view for investigating officers upon login.

```
+-----+
| [!] DEMO DATA MODE - NOT FOR OPERATIONAL USE |
+-----+
| Active Cases: 14 Open | Total Traced: ■2,500+ Cr | Alerts: 347 | [+ NEW CASE] |
+-----+
| RECENT ACTIVITY FEED |
| [10:42 AM] Wallet 0xA7F3... flagged CRITICAL (Score: 85) |
| [10:30 AM] Evidence Bank_Statement.csv hashed: SHA-256 (0x7F3A...) |
| [10:15 AM] Case CP-CYBER-2026-001 created by IO Sharma |
+-----+
```

- **Key Components:** Active Cases Panel, Quick Metric Cards, "New Case" Primary Action Button, System Audit Feed.
- **Winning Factor:** Eliminates menu navigation, providing an actionable control room feel immediately.

ORGAN 1: Case Intake Organ

Purpose: Let an investigator create or open a case.

```
+-----+
| CASE INTAKE PANEL |
+-----+
| Case ID: [ CP-CYBER-2026-001 ] (Auto-Generated) |
| FIR / NCRP Reference #: [ 1930-NCRP-2026-98124 ] |
| Victim Amount (■): [ 500000 ] Incident Date: [ 2026-09-08 10:30 ] |
| Known Wallet Address: [ 0xA7F367e92d... ] |
| Transaction Hash: [ 0x8b12f... ] Bank UTR: [ 420192830192 ] |
| Sec 94 BNSS Legal Order: [ Upload File... ] |
| Investigator Notes: [ Suspect received funds via UPI prior to USDT conversion ] |
| |
| [ PROCEED TO EVIDENCE UPLOAD ] |
+-----+
```

- **Key Components:** Auto Case ID, FIR/NCRP reference field, Victim Loss (■), Incident Timestamp, Wallet/TX Hash fields, Bank UTR field, Section 94 BNSS Order slot.
- **Why:** Acts as the mandatory entry point that mirrors standard Indian Police intake procedures.

ORGAN 2: Evidence Uploader Organ

Purpose: Ingest, parse, and cryptographic-hash case files.

```
+-----+
| EVIDENCE UPLOADER |
+-----+
| +-----+ |
| | DRAG & DROP FILES HERE | |
| | Accepts CSV, PDF, TXT only | |
| +-----+ |
| | |
| COLUMN MAPPER (Bank Statement CSV) |
| UTR Column: [ Select Column v ] | Amount Column: [ Select Column v ] |
| Timestamp: [ Select Column v ] | Bank Name: [ Select Column v ] |
| | |
| INGESTION PREVIEW & INTEGRITY HASH |
| File: Bank_Stmt_SBI.csv | Size: 1.2 MB | SHA-256: 0x9f8a7b6c5d4e3f2a1b... |
| Status: [✓] Hash Generated & Immutable Logged |
| Toggle: [X] Mark as Synthetic / Demo Data |
+-----+
```

- **Key Components:** Drag-and-drop zone, file validator, client-side SHA-256 auto-hasher, CSV column mapper dropdowns, parsing preview table.
- **Why:** Ingests unstructured bank statements from SBI, HDFC, and ICICI, securing them for court admissibility.

ORGAN 3: Graph Visualiser Organ ("The Violet Milk Map")

Purpose: Render an interactive multi-chain transaction graph.

```
+-----+
| GRAPH VISUALISER                                     |
+-----+
| (■ Victim) --Dashed Line--> (■ Bank) --Solid Line--> (■ CEX Deposit) |
|                                                         |
|                                                         | Solid Line |
|                                                         | v         |
| (■ Mixer) <--Solid Line-- (■ Wallet 0xA7F3) <--Solid Line-- (■ Bridge) |
| [CRITICAL]                ["Violet Milk"]                        |
|
| LEGEND: ■ Victim | ■ Bank | ■ Exchange | ■ Wallet | ■ Mixer | ■ Bridge |
| EDGES: --- Confirmed On-Chain - - - Inferred Correlation |
+-----+
```

- **Key Components:** Cytoscape.js canvas, color-coded node taxonomy, edge style indicators, hover metadata cards, double-click hop expander.
- **Why:** Provides a high-impact, visual trace of illicit funds moving across wallet hops.

ORGAN 4: Timeline Organ

Purpose: Display chronological fund movement.

```
+-----+
| CHRONOLOGICAL TIMELINE                             |
+-----+
| | 10:31:04 AM -- ■ Victim initiates UPI Transfer (■4,70,000) |
| | UTR: 420192830192                                         |
| | [+13 sec]                                                 |
| |-- 10:31:17 AM -- ■ Centralized Exchange Deposit Detected |
| | [+45 sec]                                                 |
| |-- 10:32:02 AM -- ■ Wallet 0xA7F3 Receives 5,200 USDT    |
| | [+3 min]                                                  |
| v 10:35:02 AM -- ■ Funds Route to Mixer Contract          |
+-----+
```

- **Key Components:** Vertical event stack, time-delta badges, category tags, cross-organ node selection sync.
- **Why:** Matches how police officers evaluate cases sequentially over time.

ORGAN 5: Risk Inspector Organ

Purpose: Explain risk scoring transparently.

```
+-----+
| RISK INSPECTOR: Wallet 0xA7F3...                     |
+-----+
| RISK SCORE: [ 65 / 100 ] - HIGH RISK                  |
|
| SCORE BREAKDOWN ("WHY?")                               |
| +20 : Proximity to known scam cluster                 |
| +15 : Rapid movement post-receipt (< 4 minutes)      |
| +15 : Mixer contract interaction                      |
| +15 : Cross-chain bridge usage                       |
|
| EVIDENCE DETAILS:                                     |
| TX Hash: 0x8b12f... | Block: #19204512 | Chain: Ethereum |
| Confidence: [✓ CONFIRMED ON-CHAIN]                    |
+-----+
```

- **Key Components:** Risk score gauge, weighted indicator breakdown, evidence hash drawer, confidence label.
- **Why:** Replaces unexplainable AI scoring with a clear, point-based audit trail suitable for court review.

ORGAN 6: Dilution Calculator Organ

Purpose: Calculate fund decay across wallet layers to eliminate false positives.

```
+-----+
| DILUTION CALCULATOR                                     |
+-----+
| INPUT: Source Wallet A (60% Illicit) sends 2 ETH       |
| DESTINATION: Wallet B contains 8 Clean ETH prior to receipt |
|
| FORMULA:                                                |
|      Dirty Received Amount                             |
| New Ratio = -----                                     |
|      Total Balance Post-Receipt                         |
|
| CALCULATION:                                            |
| Dirty Received = 2 ETH * 0.60 = 1.2 ETH                |
| New Ratio      = 1.2 ETH / (8 ETH + 2 ETH) = 12%       |
|
| RESULT: Destination Wallet Risk Adjusted from 60% -> 12% |
+-----+
```

- **Key Components:** Flow diagram, proportional haircut formula display, calculation toggle, adjusted risk gauge.
- **Why:** Demonstrates mathematical precision by ensuring clean liquidity pools are not incorrectly flagged.

ORGAN 7: Report Generator Organ (Court Dossier Exporter)

Purpose: Export a legally compliant, court-ready PDF dossier.

```
+-----+
| POLICE CASE DOSSIER GENERATOR                           |
+-----+
| PREVIEW (Page 1 of 4):                                  |
| +-----+                                              |
| | CHANDIGARH POLICE CYBER CRIME UNIT                  |
| | ELECTRONIC EVIDENCE DOSSIER (SEC 63 BSA 2023)       |
| |
| | Case Reference: CP-CYBER-2026-001      FIR / NCRP #: 1930-NCRP-2026 |
| | Target Wallet: 0xA7F367e92d...      Risk Classification: CRITICAL |
| | Document Hash: SHA-256 (0x3f2a1b...) |
| |
| +-----+                                              |
|
| [ GENERATE & DOWNLOAD DOSSIER PDF ]                    |
+-----+
```

- **Key Components:** Brief generation trigger, PDF preview modal, Section 63 BSA 2023 certificate toggle, SHA-256 verification string.
- **Why:** Converts digital analytics directly into an official paper artifact attachable to an FIR or charge sheet.

ORGAN 8: Search & Filter Organ

Purpose: Filter large node environments quickly.

```
+-----+
| SEARCH & FILTER TOOLBAR                                 |
+-----+
| Search: [ Wallet Address / TX Hash / UTR... ]         |
| Min Amount (■): [ 10000 ] Max Amount (■): [ 1000000 ] |
| Token: [ USDT v ] Chain: [ Ethereum v ] Risk Level: [ Critical & High v ] |
| View Pagination: (•) 50 Nodes ( ) 100 Nodes ( ) 200 Nodes [ RESET FILTERS ] |
+-----+
```

- **Key Components:** Multi-parameter search bar, range sliders, chain selectors, pagination toggles.
- **Why:** Maintains browser responsiveness when managing extensive networks.

ORGAN 9: Help & Limitations Organ

Purpose: Define legal boundaries and operational status.

```
+-----+
| SYSTEM NOTICE & DISCLAIMERS |
+-----+
| [!] DEMO / SYNTHETIC DATA MODE ACTIVE |
| NOTICE: This platform provides analytical leads for investigative assistance. |
| Output generated does not constitute a legal finding of guilt without |
| independent verification under Section 94 BNSS 2023. |
| |
| Authorized Pilot Support: cyber-cell-support@chandigarhpolice.gov.in |
+-----+
```

- **Key Components:** Persistent status banner, operational disclaimer, legal lead tooltip, support contacts.
- **Why:** Establishes clear legal boundaries for prototype software during police evaluations.

ORGAN 10: Audit Trail & Chain of Custody Organ

Purpose: Maintain an unalterable log of every evidence interaction as mandated by BNSS 2023.

```
+-----+
| EVIDENCE CHAIN OF CUSTODY LOG |
+-----+
| TIMESTAMP      USER ID      ACTION EXECUTED      TARGET HASH / ITEM |
| 2026-09-08 10:30:12 IO_SHARMA  File Uploaded        SHA-256: 0x7F3A... |
| 2026-09-08 10:31:05 IO_SHARMA  Graph Traversed (Hop 3) Wallet: 0xA7F3... |
| 2026-09-08 10:35:18 IO_SHARMA  Dossier PDF Exported  SHA-256: 0x3F2A... |
| |
| [ VERIFY INTEGRITY OF ALL HASHES ] [ EXPORT AUDIT TRAIL (JSON) ] |
+-----+
```

- **Key Components:** Access log table, action-to-hash binding engine, one-click hash re-verifier, audit export button.
- **Why:** Guarantees evidence integrity from initial file upload through final court report export.